

Science and Its Critics

32465

Lesson 12

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Course Summary

Review lesson before the exam – overview

- ▶ The course dealt with the present **scientific method** and the **challenges** it faces.
- ▶ This method is **logical empiricism**, i.e., it holds that the only sources of knowledge are:
 - ▶ **1. observations** and **2. logic.**
- ▶ We divided our course into three parts:
- ▶ **Part I:** Introduction to the general principles of **the scientific method**
- ▶ **Part II:** Introduction to types **of logical arguments** and how they are used in the scientific method
- ▶ **Part III: Critiques** of the scientific method (*most of the course*)

The logic of science: **theory** and **observation**

The Task		The logic
Formulation Stage	Scientific method for formulating a hypothesis - abduction	
Evaluation Stage	Scientific method for confirming a hypothesis – induction	
	Scientific method for refuting a hypothesis – deduction	
Application Stage	Scientific method for prediction and explanation an observation - deduction <i>(the D-N model)</i>	

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Recap: The logic of science: **theory** and **Theory**

The Task	The logic
Inter-Theoretic Relations: Synchronous	
Inter-Theoretic Relations: Diachronous	

Recap: The logic of science: **theory** and **Theory**

The Task		The logic
Inter-Theoretic Relations: Synchronous	Explanation of the "higher" theory by the "lower" theory, by means of bridge laws . <i>The ideal of a unity of science.</i>	
Inter-Theoretic Relations: Diachronous		

Recap: The logic of science: **theory** and **theory**

	The Task	The logic
Inter-Theoretic Relations: Synchronous	Explanation of the "higher" theory by the "lower" theory, by means of bridge laws (deduction). <i>The ideal of the unity of science.</i>	1. H1 (e.g., physics) 2. Bridge Laws 3. Therefore, H2 (e.g., chemistry)
Inter-Theoretic Relations: Diachronous		

Recap: The logic of science: **theory** and **theory**

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Inter-Theoretic Relations: Diachronous	A. When a theory is refuted (deduction): Simply formulate a new theory in its place – there is no direct relationship between the old and new theories. (abduction)	

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Inter-Theoretic Relations: Diachronous	A. When a theory is refuted (deduction): Simply formulate a new theory in its place – there is no direct relationship between the old and new theories. (abduction)	1. If H1 then not e1 2. e1 3. Therefore, not 1H 1. e1 2. The best explanation for e1 is H2 3. Therefore, H2

Recap: The logic of science: **theory** and **theory**

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Inter-Theoretic Relations: Diachronous	B. <u>When there are competing theories:</u> Decisive experiment (deduction ; Induction)	
	C. <u>When we want to propose a better theory:</u> Explanation of the old theory by the new theory (Deduction) <i>The ideal of accumulative science</i>	

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	C. <u>When we want to propose a better theory:</u> Explanation of the old theory by the new theory (Deduction) <i>The ideal of a accumulative science</i>	1. H1 (e.g., Einstein's relative velocity) 2. Therefore, H*2 (e.g., Newtonian relative velocity)

Recap: The logic of the scientific method - summary

	Formulation of theory	Confirming a theory	Refutation of a theory	Explanation and prediction of observations	Explanation of an old theory by a new theory (diachronically)	Explanation of a theory by a more basic theory (synchronous)
Assumptions	Observations	Observations	Observations	Theory (=law of nature); initial conditions	Theory (new)	Theory (basic), Bridging statements
Rule of inference	Abduction	Induction	Deduction	Deduction	Deduction	Deduction
Conclusion	Theory	Theory	Theory	Observations	Theory (old)	Theory (superior)

- ▶ The table summarizes the key principles of the positivist **scientific method**.
- ▶ This method serves as guiding instructions on how to carry out all major tasks of science using only logic and observations.
- ▶ On the face of it, such 'algorithm'; can therefore be uploaded to a computer, which will then be able to conduct research that meets scientific standards. However, there are **challenges to each and every step**

Critiques of the scientific method: **theory** and **observation**

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Evaluation Stage	Scientific method for confirming a hypothesis – induction	1. If H, then e 2. e 3. Therefore, (probably) H
	Scientific method for refuting a hypothesis – deduction	1. If H, then e 2. Not-e 3. Therefore, not H
Application Stage	Scientific method for prediction and explanation an observation - deduction <i>(the D-N model)</i>	1. H 2. Initial conditions I 3. Therefore, e

Critiques of the scientific method: **theory** and **observation**

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Formulation Stage	Scientific method for formulating a hypothesis - abduction	Not exactly a method
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Application Stage	Scientific method for prediction and explanation an observation - deduction <i>(the D-N model)</i>	1. H 2. Initial conditions I 3. Therefore, e

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Application Stage	Scientific method for prediction and explanation an observation - deduction <i>(the D-N model)</i>	Natural laws vs. accidental generalisations

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Application Stage	Scientific method for prediction and explanation an observation - deduction (<i>the D-N model</i>)	Natural laws vs. accidental generalisations
	Scientific method for explanation - deduction (<i>the D-N model</i>)	The model is both too weak and too strong

Critiques of the scientific method: **theory** and **theory**

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Inter-Theoretic Relations: Diachronous	A. When a theory is refuted (deduction): Simply formulate a new theory in its place – there is no direct relationship between the old and new theories. (abduction)	1. If H1 then <u>not</u> e1 2. e1 3. Therefore, not H1 1. N e1 2. The best explanation for e1 is H2 3. Therefore, H2

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Inter-Theoretic Relations: Synchronous	Explanation of the "higher" theory by the "lower" theory, by means of bridge laws (deduction). <i>The ideal of the unity of science.</i>	A. Especially in the social sciences: 1. Multiple realizability 2. Accessibility 3. Consciousness (Searle) 4. Experiences (Jackson) B. <u>Kuhn</u> : meaning is paradigm-dependent
Inter-Theoretic Relations: Diachronic	<u>A. When a theory is refuted</u> (deduction): Simply formulate a new theory in its place – there is no direct relationship between the old and new theories. (abduction)	1. If H1 then <u>not</u> e1 2. e1 3. Therefore, not H1 1. N e1 2. The best explanation for e1 is H2 3. Therefore, H2

Critiques of the scientific method: **theory** and **theory**

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Inter-Theoretic Relations: Diachronous	<u>A. When a theory is refuted</u> (deduction): Simply formulate a new theory in its place – there is no direct relationship between the old and new theories. (abduction)	<u>Kuhn</u> : 1. This is not worthwhile (and in practice is not done), because first the existing theory must be exhausted. 2. 'Kuhn's loss'

Recap: The logic of science: **theory** and **theory**

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	The Task	Challenge
Inter-Theoretic Relations: Diachronous	B. <u>When there are competing theories:</u> Decisive experiment (deduction; Induction)	<u>Kuhn</u> : Methodology is paradigm-dependent . So The rules of evaluation are part of the paradigm. Therefore, theories cannot be compared.
	C. <u>When we want to propose a better theory:</u> Explanation of the old theory by the new theory (Deduction) <i>The ideal of a accumulative science</i>	1. H1 (e.g., Einstein's relative velocity) 2. Therefore, H*2 (e.g., Newtonian relative velocity)

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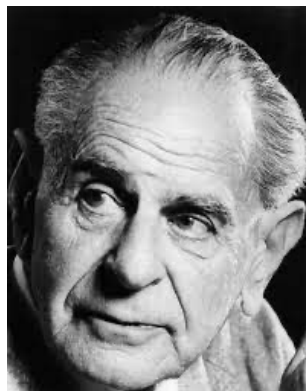
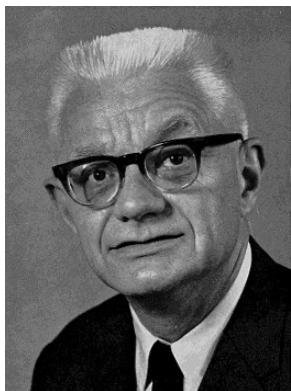
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Inter-Theoretic Relations: Diachronous	B. <u>When there are competing theories:</u> Decisive experiment (deduction; Induction)	<u>Kuhn</u> : Methodology is paradigm-dependent. So The rules of evaluation are part of the paradigm. Therefore, theories cannot be compared.
	C. <u>When we want to propose a better theory:</u> Explanation of the old theory by the new theory (Deduction) <i>The ideal of a accumulative science</i>	<u>Kuhn</u> : semantic holism. Meanings are paradigm-dependent. Hence, translation is impossible.

Recap: The logic of science: **theory** and **theory**

- ▶ In addition to all this, Kuhn also argues for a **conceptual** lack of common standards, i.e. **observation is paradigm-dependent**, which leads to **the absence of neutral observations**.
- ▶ The implication: every scientific ideal regarding the relationship between **observation and hypothesis/theory** also collapses. (Regardless of all the critiques we have learned about this relationship: the raven paradox, Duhem-Quine, etc.)
- ▶ It follows, then, that Kuhn's position undermines the **entire positivist scientific method!**
- ▶ The problem: this is the best method we have.
- ▶ How do we continue to do science?

How do we proceed?

- ▶ In the absence of better scientific method, we continue according to the positivist method, while **being aware** of all the difficulties and insights.
- ▶ For example A: When encountering an **observation** that does **not fit a hypothesis**, we ask whether it actually refutes the hypothesis, or perhaps only some of our **background assumptions** (e.g., regarding the measuring instruments).
- ▶ Example B: We become sensitive to the way we **describe** the observation, while being aware that every description is already **loaded** with theoretical assumptions (depending on the paradigm).
- ▶ **Etc.**
- ▶ **And most importantly, we should bear in mind that science is by far still the best tool we have for explaining reality, predicting reality, and producing new technologies!**



► **I really enjoyed teaching you – thank you very much!**

► Welcome to stay in touch:

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